

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B.Tech Examination (CE / IT)

December 2012

CE403: Data Warehousing & Data Mining

Date: 06.12.2012, Thursday

Time: 10:00 am to 1:00 pm

Maximum Marks:70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION-I

Q-1

- (a) How is a *data warehouse* different from a database? How are they similar? 2
- (b) Describe why concept hierarchies are useful in data mining. 2
- (c) State why, for the integration of multiple heterogeneous information sources, many companies in industry prefer the update-driven approach (which constructs and uses data warehouses), rather than the query-driven approach (which applies wrappers and integrators). Describe situations where the query-driven approach is preferable over the update-driven approach. 3

Q-2

- (a) Describe the steps involved in data mining when viewed as a process of knowledge discovery(KDD). 5
- (b) Use the methods below to normalize the following group of data: 3
200, 300, 400, 600, 1000
(1) min-max normalization by setting min = 0 and max = 1
(2) z-score normalization
(3) normalization by decimal scaling
- (c) In data warehouse technology, a multiple dimensional view can be implemented by a relational database technique (*ROLAP*), or by a multidimensional database technique (*MOLAP*), or by a hybrid database technique (*HOLAP*). Briefly describe each implementation technique. 6

OR

Q-2

- (a) List and describe the five primitives for specifying a data mining task. 5
- (b) Explain the three tier architecture of data warehouse with neat sketch. 3
- (c) Explain the following heuristics methods for attribute subset selection with neat figure: 6
 1. Step-wise forward selection
 2. Step-wise backward elimination

Q-3 Attempt Any Two.

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- (a) Suppose that a data warehouse consists of the four dimensions, date, spectator, location, and game, and the two measures, count and charge, where charge is the fare that a spectator pays when watching a game on a given date. Spectators may be students, adults, or seniors, with each category having its own charge rate.
 - (1) Draw a star schema diagram for the data warehouse.

- (2) Starting with the base cuboid [date, spectator, location, game], what specific OLAP operations should one perform in order to list the total charge paid by student spectators at GM Place in 2010?
- (b) Why sampling can be used as a data reduction technique? Explain with example Sampling with replacement, Cluster Sampling and Stratified Sampling.
- (c) Suppose that the data for analysis includes the attribute *age*. The *age* values for the data tuples are (in increasing order) 35, 25, 16, 16, 20, 40, 22, 22, 45, 25, 25, 25, 30, 33, 20, 35, 35, 35, 36, 46, 19, 52, 70, 13, 15, 21, 33.
1. Partition them into three bins by each of the following methods.
 - (a) equal-frequency (equidepth) partitioning
 - (b) equal-width partitioning
 2. Use *smoothing by bin means*, to smooth the above data with bin depth of 3. Illustrate your steps. Comment on the effect of this technique for the given data.

SECTION-II

Q-4

- (a) What is the difference between clustering and classification? how are they similar? 2
- (o) What is the limitation of accuracy as a classifier performance measure? What other measures can be used in place of accuracy? 2
- (c) How can we mine multilevel association rules efficiently using concept hierarchies? 3

Q-5

- (a) Briefly outline the k-means partitioning algorithm for clustering. 3
- (b) Explain the basic algorithm for attribute-oriented induction using a database approach. 5
- (c) Define k-itemset. Explain the join and prune steps and the terminating condition of Apriori algorithm. 6

OR

Q-5

- (a) Briefly outline the major steps of decision tree classification. 3
- (b) What is Bayesian Belief Networks? Explain the component by which Bayesian belief network is defined. 5
- (c) Explain following Issues regarding classification and prediction with their significance. 6
- 1.Data Cleaning 2.Relevance Analysis 3.Data Transformation

Q-6 Attempt Any Two.

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- (a) The following contingency table summarizes supermarket transaction data, where A refers to the transactions containing A, ~A refers to the transactions that do not contain A, B refers to the transactions containing B, and ~B refers to the transactions that do not contain B.

	A	~A
B	2000	500
~B	1000	1500

- (1) Suppose that the association rule " $A \Rightarrow B$ " is mined. Given a minimum support threshold of 25% and a minimum confidence threshold of 50%, is this association rule strong?
- (2) Based on the given data, is the purchase of A independent of the purchase of B? If not, what kind of *correlation* relationship exists between the two?

- (b) The following table consists of training data from an student database. The data have been generalized. For example, "30....40" for *age* represents the age range of 30 to 40. Let *buys_computer* be the class label attribute.

Given a test tuple having the values $X = (\text{age} = "<30", \text{income} = \text{medium}, \text{student} = \text{yes}, \text{credit rating} = \text{fair})$. What would a naive Bayesian classification for the tuple be?

rid	age	income	student	credit_rating	Class: buys_computer
1	<30	high	no	fair	no
2	<30	high	no	excellent	no
3	30-40	high	no	fair	yes
4	>40	medium	no	fair	yes
5	>40	low	yes	fair	yes
6	>40	low	yes	excellent	no
7	30-40	low	yes	excellent	yes
8	<30	medium	no	fair	no
9	<30	low	yes	fair	yes
10	>40	medium	yes	fair	yes
11	<30	medium	yes	excellent	yes
12	30-40	medium	no	excellent	yes
13	30-40	high	yes	fair	yes
14	>40	medium	no	excellent	no

- (c) A database has 5 transactions. Let *minimum support* = 60% and *minimum confidence* = 80%.

TID	items bought
T1	{B, F, G, H, J}
T2	{K, F, G, H, I, J}
T3	{B, M, H, I}
T4	{B, N, O, H, J}
T5	{C, F, F, H, P, I}

Find all frequent item sets using Apriori and list all *strong* association rules with given minimum support and confidence
